



PPMI Complete Multimodal Feature Dictionary

Cohort: De Novo PD & Healthy Controls (N = 9,270)

Dataset: PPMI Longitudinal (5.67 GiB)

Master Dataset Specifications & Variable Mappings for Parkinson's Progression Capstone

★ PRIMARY PREDICTION TARGET & MEDICATION CONFOUNDER

NP3TOT

 — MDS-UPDRS Part III Total Motor Score (0 – 132)

Predicting exact motor score at **t + 12 months** and **t + 24 months** into the future.

Medication Confounder:

PDSTATE

 (ON vs. OFF)

Medication Dose:

LEDD

 (Levodopa Daily mg)

1. Master Identifiers & Temporal Indexing

VARIABLE NAME	SOURCE FILE	TYPE	CLINICAL / MACHINE LEARNING ROLE
PATNO	All CSV tables	KEY	Unique patient identifier. Master key for relational joins and patient-level split (70/15/15).
EVENT_ID	All CSV tables	CATEGORICAL	Clinical visit identifier (BL =Baseline, V04 =Year 1, V06 =Year 2, V08 =Year 3).
INFODT	All CSV tables	DATE	Date of clinical check-up. Used to compute true elapsed days (Δt) for GRU-D decay.
PDSTATE	MDS-UPDRS_Part_III...	BINARY	Medication state during motor exam (ON vs. OFF medication). Essential covariate.

2. Modality 1A: In-Clinic Motor Exam Subscores MOTOR TIME-SERIES

VARIABLE NAME	SOURCE FILE	TYPE	CLINICAL MOTOR EXAMINATION ITEM
NP3SPCH , NP3FACXP	MDS-UPDRS_Part_III...	0 – 4	Speech impairment and facial expression masking (hypomimia).
NP3RIGN , NP3RIGRU/LU	MDS-UPDRS_Part_III...	0 – 4 EACH	Muscle Rigidity in neck, right/left arms, and right/left legs (NP3RIGRL/LL).
NP3FTAPR , NP3FTAPL	MDS-UPDRS_Part_III...	0 – 4 EACH	Finger Tapping: speed, amplitude, and hesitations for right and left hands.
NP3HMOVR/L , PRSPR/L	MDS-UPDRS_Part_III...	0 – 4 EACH	Rapid alternating hand movements (fist opening/closing and pronation/supination).
NP3TTAPR/L , LGAGR/L	MDS-UPDRS_Part_III...	0 – 4 EACH	Toe tapping agility and leg agility (testing lower limb bradykinesia).
NP3RISNG	MDS-UPDRS_Part_III...	0 – 4	Arising from a chair with arms crossed over chest.
NP3GAIT , NP3FRZGT	MDS-UPDRS_Part_III...	0 – 4 EACH	Walking stride length, arm swing, shuffling, and sudden Freezing of Gait episodes.
NP3PSTBL , NP3POSTR	MDS-UPDRS_Part_III...	0 – 4 EACH	Postural stability (pull test recovery) and stooped posture severity.
NP3BRADY	MDS-UPDRS_Part_III...	0 – 4	Global body bradykinesia (overall slowness and lack of spontaneous movement).
NP3PTRMR/L , RTMR...	MDS-UPDRS_Part_III...	0 – 4 EACH	Action/postural tremor of hands, and resting tremor of arms, legs, and lip/jaw.

3. Modality 1B: Daily Living & Non-Motor Clinical Domains COGNITION, SLEEP, AUTONOMIC

VARIABLE NAME	SOURCE FILE	TYPE	CLINICAL DOMAIN & SIGNIFICANCE
NP1RTOT	MDS-UPDRS_Part_I...	CONTINUOUS	Non-motor aspects of daily living (cognitive impairment, hallucinations, apathy).
NP2PTOT	MDS_UPDRS_Part_II...	CONTINUOUS	Motor aspects of daily living (swallowing, dressing, hygiene, writing, turning in bed).

VARIABLE NAME	SOURCE FILE	TYPE	CLINICAL DOMAIN & SIGNIFICANCE
NP4WDYSK, NP4OFF	MDS-UPDRS_Part_IV...	CONTINUOUS	Motor complications: daily duration of involuntary dyskinesia and "OFF" state time.
MSEADLG	Modified_Schwab...	0 - 100%	Schwab & England Activities of Daily Living: measure of functional independence.
MCATOT	Montreal_Cognitive...	0 - 30	MoCA Total Score: Global cognitive decline, executive function, visuospatial memory.
SCAU1 - SCAU25	SCOPA-AUT_26Sep2026.csv	COMPOSITE	Autonomic dysfunction: gastrointestinal, urinary, orthostatic, and cardiovascular issues.
GDSTOT	Geriatric_Depression...	0 - 15	Geriatric Depression Scale: tracks depression and mood decline in Parkinson's.
DRMVIVID, SLPLMBMV	REM_Sleep_Behavior...	BINARY	RBD sleep questionnaire: acting out violent dreams, thrashing (strong early PD marker).
ESS_TOT	Epworth_Sleepiness...	0 - 24	Epworth Sleepiness Scale: excessive daytime somnolence.
UPSIT_TOTAL	University_of_Penn_UPSIT	0 - 40	Smell identification test: olfactory loss is present in >90% of early PD patients.

4. Demographics, Genetic Risk & Medication Confounders

STATIC & CONFOUNDERS

VARIABLE NAME	SOURCE FILE	TYPE	CLINICAL / MACHINE LEARNING ROLE
AGE_AT_VISIT	Age_at_visit...	YEARS	Patient age at time of clinical exam. Strongest natural risk factor for progression.
SEX, EDUCYRS	Demographics_26Sep2026.csv	CAT / INT	Biological sex (higher incidence in males) and years of education (cognitive reserve).
LEDD	LEDD_Concomitant_Med...	MG / DAY	Levodopa Equivalent Daily Dose: Total daily dopaminergic medication load.
LRRK2, GBA, APOE	iu_genetic_consensus...	BINARY	Carrier flags for pathogenic Parkinson's mutations and APOE ε4 Alzheimer's allele.
BIODADPD, BIOMOMPD	Family_History_26Sep2026.csv	BINARY	Family history of Parkinson's disease in first-degree relatives (father, mother, sibling).

5. Modality 2: Biofluid / Biospecimen Biomarkers

CSF & BLOOD ASSAYS

VARIABLE NAME	SOURCE FILE	TYPE	BIOLOGICAL MEASUREMENT & TARGET MECHANISM
ABeta42 (Aβ1-42)	Current_Biospecimen...	PG / ML	Cerebrospinal fluid Amyloid-Beta: lower CSF levels correlate with cognitive decline.
tTau, pTau181	Current_Biospecimen...	PG / ML	CSF Total Tau and Phospho-Tau: indicators of ongoing neuronal death and axonal injury.
Alpha-synuclein	Current_Biospecimen...	PG / ML	Total alpha-synuclein protein concentration in lumbar puncture CSF.
SAA_Status	SAA_Biospecimen...	POS / NEG	Seed Amplification Assay: Direct biochemical detection of pathological α-syn seeds.
Fmax, T50	SAA_Biospecimen...	KINETICS	SAA Aggregation kinetics: maximum fluorescence intensity and time to 50% aggregation.

6. Modality 3: Imaging-Derived Features (Pre-computed)

DATSCAN SPECT & STRUCTURAL MRI

VARIABLE NAME	SOURCE FILE	TYPE	NEUROIMAGING MEASUREMENT & BRAIN STRUCTURE
PUTAMEN_L/R_REF_CWM	Xing_Core_Lab_Quant_SBR	SBR RATIO	Putamen Striatal Binding Ratio: Directly measures dopamine terminal loss (hallmark of PD).
CAUDATE_L/R_REF_CWM	Xing_Core_Lab_Quant_SBR	SBR RATIO	Caudate Striatal Binding Ratio: dopamine binding in cognitive/associative basal ganglia.

VARIABLE NAME	SOURCE FILE	TYPE	NEUROIMAGING MEASUREMENT & BRAIN STRUCTURE
Striatal Asymmetry	Computed Feature	INDEX	Asymmetry between left and right putamen binding ($ L - R / (L + R)$).
Hippocampus_L/R	FS7_ASEG_VOL_26Sep2026	MM ³	FreeSurfer 7 Left and Right Hippocampal volume: marker of cognitive neurodegeneration.
Lateral_Ventricles	FS7_ASEG_VOL_26Sep2026	MM ³	Ventricular enlargement: global cerebral atrophy marker (identified by Dentamaro et al.).
Brain_Stem	FS7_ASEG_VOL_26Sep2026	MM ³	Brainstem volume (where the dopamine-producing Substantia Nigra is located).
eTIV	FS7_ASEG_VOL_26Sep2026	MM ³	Estimated Total Intra-Cranial Volume: Used to normalize brain region sizes across heads.
Cortical Thickness	FS7_APARC_CTH_26Sep2026	MM	Mean cortical thickness in temporal and prefrontal lobes (linked to executive dysfunction).